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SCHOOL OF CHEMICAL AND BIOTECHNOLOGY · DEPARTMENT OF BIOINFORMATICS

MINI RESEARCH PROPOSAL · BIG DATA FOR A BIOLOGICAL PROBLEM

NAME	Sreya C V	REGISTER NO.	127013054
PROGRAMME	B.Tech. Bioinformatics, Year IV, Sem 7	COURSE	Big Data Analytics
TEAM MEMBERS		REGISTER NO.	
		REGISTER NO.	
		REGISTER NO.	
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1. Title

Ranking approved drugs for polycystic ovary syndrome by network proximity analysis of drug-target and protein-interaction data using Apache Spark

2. The Biological Problem

Polycystic ovary syndrome (PCOS) is the most common endocrine disorder in women of reproductive age, with a pooled prevalence of roughly 10% (Bozdag et al., 2016). It acts through three interacting axes: insulin resistance, hyperandrogenism and chronic low-grade inflammation (Escobar-Morreale, 2018), and it raises lifetime risk of type 2 diabetes, cardiovascular disease and endometrial cancer. Despite this burden, no drug carries a regulatory approval for PCOS. The 2023 international guideline recommends metformin (a diabetes drug), spironolactone (an antihypertensive), combined oral contraceptives, and letrozole or clomiphene for ovulation induction (Teede et al., 2023). Every one of these is borrowed and symptomatic.

What is unsolved. Each borrowed drug was found individually, from clinical observation. There has never been a systematic, evidence-weighted screen of the entire approved pharmacopoeia against PCOS biology. De novo drug development takes 10 to 15 years and over one billion US dollars (DiMasi et al., 2016), whereas an approved drug already has human safety and pharmacokinetic data and can reach a trial far faster (Pushpakom et al., 2018). A data-driven repurposing screen is therefore the quickest route to new PCOS treatment options.

Promised output. A ranked list of about 2,800 approved drugs scored by how close their protein targets sit to the PCOS gene module in the human interactome, with the evidence path shown for each candidate, validated against drugs already used for PCOS.

3. Why This Is a Big Data Problem

The PCOS gene set itself is small, about 20 genome-wide significant loci (Day et al., 2018) and a few hundred Open Targets associations. The scale comes from the drug side and the network side.

- **Volume.** ChEMBL holds 20 million bioactivity measurements across 2.4 million compounds and 15,000 targets, about 25 GB loaded. STRING human has 12 million scored protein links. Open Targets distributes 18 million evidence strings, about 50 GB in Parquet.
- **Variety.** Relational bioactivity tables, disease-evidence JSON, a weighted graph, drug-annotation XML and clinical-trial records, each with its own identifier system (ChEMBL, UniProt, Ensembl, STRING, EFO).

- **Veracity.** Experimental versus predicted interactions, evidence scores on a 0 to 1 scale, and sources with restrictive licences (KEGG, DisGeNET bulk) that must be excluded and substituted.
- **Velocity.** ChEMBL and Open Targets release quarterly, so the pipeline must be reproducible and re-runnable rather than a one-off script.

The step an ordinary laptop cannot do. Network proximity must be computed for each of about 2,800 drugs on a 19,600-node, 12-million-edge graph, and each drug's score compared with 1,000 degree-matched random target sets: roughly 2.8 million shortest-path computations. The graph plus per-drug state exceeds 8 GB of RAM, and one core would need days. The task is embarrassingly parallel across drugs, so it maps directly onto a Spark map over drug partitions with the graph as a broadcast variable. Spark is chosen over Hadoop MapReduce because the computation is iterative and Spark keeps intermediate state in memory instead of writing to disk each round (Zaharia et al., 2016; Dean & Ghemawat, 2008). Open Targets already ships Parquet, Spark's native columnar format.

4. Data

Source	Type and role	Approximate size, format	One record looks like	Licence
ChEMBL v35 (Zdrazil et al., 2023)	Drug-target bioactivities	20 M activities, 15 k targets; ~4 GB compressed, Parquet	CHEMBL25 (aspirin) → PTGS1, IC50 = 32 nM, pchembl 7.5	CC BY-SA
DrugBank (Knox et al., 2023)	Approval status, indications	16.5 k drugs, 2.8 k approved; XML 1.5 GB	DB00331 metformin, approved, target PRKAB1, indication T2D	Academic
Open Targets Platform (Buniello et al., 2024)	PCOS-gene associations with evidence scores	18 M evidence strings; associations slice ~2 GB Parquet	ENSG00000105329 (TGFB1), EFO_0000660, score 0.42	CC0
GWAS Catalog (Sollis et al., 2022)	Genetic seed genes	700 k associations; TSV ~300 MB	rs11031006, FSHB, PCOS, $p = 1 \times 10^{-12}$	Open
STRING v12, human (Szklarczyk et al., 2022)	Protein-interaction network	19.6 k proteins, 12 M links; TSV	9606.ENSP00000284981 – 9606.ENSP00000262607, score 912	CC BY 4.0
UniProt / Ensembl (The UniProt Consortium, 2025)	Identifier bridge	20 k human proteins; mapping files, hundreds of MB	P37231 ↔ ENSG00000132170 ↔ PPARG	CC BY 4.0
ClinicalTrials.gov (NLM)	Validation set	500 k trials; JSON/CSV	NCT id, metformin, condition PCOS, phase 3	Open

Table 1. Public data sources with role, size, one example record and licence.

KEGG bulk download is licensed, so Reactome is used for pathway context. DisGeNET now restricts bulk access, so the disease side is built on Open Targets and DisGeNET is cited only as a cross-check.

5. Approach

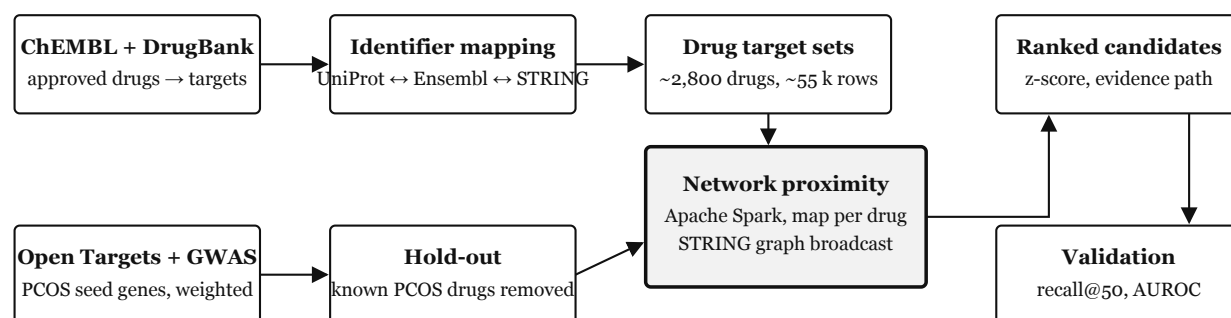


Figure 1. Pipeline. The shaded step is the distributed computation; every other step produces a small output.

1. **Ingest and store.** Download ChEMBL, DrugBank, Open Targets associations, GWAS Catalog and STRING; convert everything to Parquet in HDFS or local object storage so Spark reads only the columns each step needs.
2. **Approved drug set.** ChEMBL molecules with `max_phase = 4`, cross-checked with the DrugBank "approved" flag. About 2,800 drugs.
3. **Drug target sets.** Spark join over 20 million ChEMBL activity rows, keeping human single-protein targets with `pchembl ≥ 6` (potency $\leq 1 \mu\text{M}$). Output: drug $\rightarrow$ set of UniProt accessions, about 55,000 rows. *Big data method: distributed join.*
4. **PCOS seed set.** Open Targets associations for `EF0_0000660` with score ≥ 0.1 , plus GWAS Catalog mapped genes, weighted by evidence score. 300 to 500 genes.
5. **Identifier mapping.** UniProt $\leftrightarrow$ Ensembl $\leftrightarrow$ STRING. Every unmapped identifier is logged and the loss rate reported.
6. **Hold-out.** Before scoring, drugs already used or in trials for PCOS (metformin, spironolactone, letrozole, clomiphene, pioglitazone, liraglutide, statins) are removed from seed construction and kept as the validation set.
7. **Network proximity** (Guney et al., 2016). For each drug, the mean shortest-path distance from its targets to the seed set on STRING (score ≥ 400), compared with 1,000 degree-matched random target sets, giving a z-score. *Big data method: Spark map over drug partitions, graph broadcast to every executor.* This is the step that needs distribution.
8. **Rank and annotate.** Sort by z-score; attach current indication, closest PCOS genes and the evidence type behind each link.
9. **Validate.** Recall of the held-out PCOS drugs in the top 50 and AUROC against all other approved drugs, following Cheng et al. (2018).

6. Expected Outcome

Product. A ranked table of approved drugs with proximity z-score, current indication, nearest PCOS genes and evidence type; a network figure of the top ten candidates; and a reproducible Spark pipeline that can be re-run on each quarterly data release.

Measure of success. At least four of the roughly eight held-out PCOS drugs appear in the top 50 of about 2,800 (fewer than 0.2 expected by chance), with AUROC above 0.7. If the hold-out is not recovered, the negative result still shows whether PCOS biology is reachable through the currently druggable target space, which is itself informative.

7. Challenges and Limitations

- **Hardest part: identifier mapping** across five sources with different ID systems. A 5 to 10% loss of targets is typical and will be measured and reported, not hidden.
- **Small, noisy disease gene set.** Seeds are weighted by evidence type, and an ablation tests whether genetic seeds alone recover the hold-out better than all seeds.
- **Literature bias.** Well-studied proteins have more edges, so raw proximity favours them. Degree-matched randomisation controls for this.
- **Data leakage.** If known PCOS drugs stayed in seed construction the validation would be circular, so they are held out first.
- **Honest limitation.** The output is a computational prioritisation only. Any candidate needs experimental and clinical validation before it means anything for patients.

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Signature of Student

Signature of Course Faculty